

Small Oscillations (Goldstein Ch. 6) — approved segments

27 approved segments, in reading order

SMALL OSCILLATIONS (Chapter 6 Goldstein)

1D CASE: $\ddot{q} = f(q)$

EQUILIBRIUM $\Leftrightarrow 0 = f(q_0)$
at $q = q_0$

Now we want to know what happens if we move a little bit away from q_0

$q = q_0 + \eta$ η "SMALL"

$(q_0 + \eta)'' = \ddot{\eta}$

$f(q_0 + \eta) = \cancel{f(q_0)} + \eta f'(q_0) + \dots$

$\ddot{\eta} \approx f'(q_0) \eta$

SMALL DEVIATION

(EQUATION OF MOTION IS **LINEAR** IN SMALL DEVIATIONS)

OSCILLATIONS $\Leftrightarrow f'(q_0) < 0$

INSTABILITY $\Leftrightarrow f'(q_0) > 0$

$\omega = \sqrt{-f'(q_0)}$

$\lambda = \sqrt{f'(q_0)}$

$\eta = \eta_0 \cos(\omega t + \varphi_0)$

$\eta = A e^{\lambda t} + B e^{-\lambda t}$

More than one degree of freedom

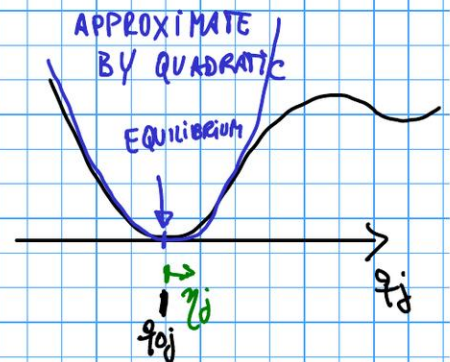
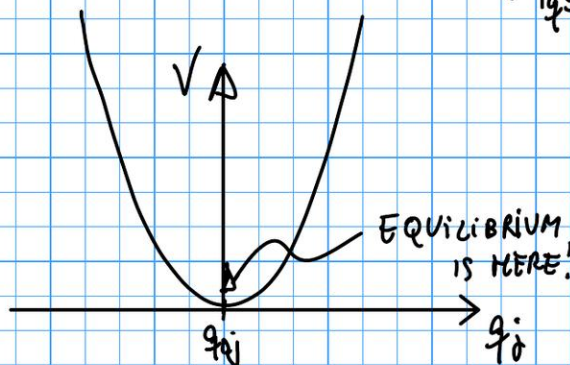
(Simplify by assuming $V = V(q_1, \dots, q_n, t)$)

$L = T - V$

Equilibrium. $\rightarrow$ GENERALIZED Q. - $\partial V / \partial q_i = 0$

$$L = T - V$$

Equilibrium at $(q_1, \dots, q_n) = (q_{01}, \dots, q_{0n}) \Leftrightarrow$ GENERALIZED FORCES $Q_j = - \left. \frac{\partial V}{\partial q_j} \right|_{\vec{q} = \vec{q}_0} = 0$



TAYLOR EXPAND V
TO 2nd ORDER IN η

$$V(q_1, \dots, q_n) = V(q_{01}, \dots, q_{0n}) + \sum_{j=1}^n \left. \frac{\partial V}{\partial q_j} \right|_{\vec{q}_0} \eta_j + \frac{1}{2} \sum_{i,j=1}^n \left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\vec{q}_0} \eta_i \eta_j + \dots$$

$q_i = q_{0i} + \eta_i$

TAYLOR EXPAND V
TO 2nd ORDER IN η

$$V(q_1, \dots, q_n) = V(q_{01}, \dots, q_{0n}) + \underbrace{\sum_{j=1}^n \frac{\partial V}{\partial q_j} \bigg|_{\vec{q}_0}}_{\substack{\text{CONSTANT} \\ \Rightarrow \text{NO WORRIES}}} \eta_j + \frac{1}{2} \sum_{i,j=1}^n \underbrace{\frac{\partial^2 V}{\partial q_i \partial q_j} \bigg|_{\vec{q}_0}}_{\substack{= V_{ij} \\ \text{m} \times \text{n MATRIX} \\ V_{ij} = V_{ji}}} \eta_i \eta_j + \dots$$

$-Q_j = 0$ $\dot{q}_{0j}$

$$q_j = q_{0j} + \eta_j$$

$$T = \frac{1}{2} \sum_{i,j=1}^n \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j} \bigg|_{\substack{\dot{q}_i=0 \\ \dot{q}_j=0}} \dot{\eta}_i \dot{\eta}_j$$

$T_{ij} = T_{ji}$ T_{ij} $\dot{q}_i = 0, \dot{q}_j = 0$

$$T = \sum_{p=1}^N \frac{m_p}{2} \left(\dot{\vec{r}}_p \right)^2$$

ASSUME

$$\dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$$

$$L = \frac{1}{2} \sum_{i,j=1}^n (T_{ij} \dot{\eta}_i \dot{\eta}_j - V_{ij} \eta_i \eta_j)$$

$$\frac{\partial}{\partial \eta_2} \left(\frac{1}{2} [V_{11} \eta_1 \eta_1 + V_{12} \eta_1 \eta_2 + V_{21} \eta_2 \eta_1 + V_{22} \eta_2 \eta_2] \right) = \frac{1}{2} [(V_{12} + V_{21}) \eta_1 + 2V_{22} \eta_2] \xrightarrow{V_{12}=V_{21}} V_{21} \eta_1 + V_{22} \eta_2$$

($\frac{1}{2}$ PREFACTOR CANCELS WITH FACTOR OF 2 IN THE DERIVATIVE)

In general:

$$\frac{\partial L}{\partial \eta_i} = - \sum_{j=1}^n V_{ij} \eta_j$$

$$p_i = \frac{\partial L}{\partial \dot{\eta}_i} = \sum_{j=1}^n T_{ij} \dot{\eta}_j$$

SIMPLE CASE $T = \frac{m}{2} \dot{\eta}^2$ $V = \frac{k}{2} \eta^2$
 $L = \frac{m}{2} \dot{\eta}^2 - \frac{k}{2} \eta^2$ $\frac{\partial L}{\partial \eta} = -k\eta$ $\frac{\partial L}{\partial \dot{\eta}} = m\dot{\eta}$
 $0 = \frac{\partial L}{\partial \eta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}} \right) = -k\eta - m\ddot{\eta} \Rightarrow \boxed{m\ddot{\eta} = -k\eta}$

Laprange
Eqs:

$$0 = \frac{\partial L}{\partial \eta_i} - \dot{p}_i \Rightarrow$$

$$0 = \sum_{j=1}^n -V_{ij} \eta_j - \sum_{j=1}^n T_{ij} \ddot{\eta}_j$$

$i=1, \dots, n$

Propose: $\eta_i = \eta_{i0} \cos(\omega t + \delta_i) \Rightarrow \ddot{\eta}_i = -\omega^2 \eta_i$ $i=1, \dots, n$

eqns: $\sum_{j=1}^n \ddot{\eta}_j = 0$ and $\sum_{j=1}^n \eta_j = 0$

Propose: $\eta_j = \eta_{j0} \cos(\omega t + \delta_j) \Rightarrow \ddot{\eta}_j = -\omega^2 \eta_j \quad i=1, \dots, n$

$$0 = \sum_{j=1}^n (V_{ij} \eta_j - \omega^2 T_{ij} \eta_j) = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) \eta_{j0} \cos(\omega t + \delta_j) \quad i=1, \dots, n$$

BUT ALSO, since $\cos(\omega t + \delta_j) \neq 0$ for most times:

BUT ALSO, since $\cos(\omega t + \phi_i) \neq 0$ for most times:

$$0 = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) \eta_{j0} \quad i=1, \dots, n$$

EIGENVALUE

EIGENVECTOR

ω : NORMAL
MODE FREQUENCY

η_{j0} : NORMAL
MODE

For the η_{j0} to be nontrivial, we need

$$\text{Det}[V_{ij} - \omega^2 T_{ij}] = 0$$

ASIDE:

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More details on the Kinetic Energy part:

$$T = \sum_{p=1}^N \frac{m_p}{2} (\dot{\vec{r}}_p)^2 = \sum_{p=1}^N \frac{m_p}{2} \left(\sum_{i=1}^n \frac{\partial \vec{r}_p}{\partial q_i} \dot{q}_i + \frac{\partial \vec{r}_p}{\partial t} \right) \cdot \left(\sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t} \right)$$

$$\vec{r}_p = \vec{r}_p(q_1, \dots, q_n, t) \Rightarrow \dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$$

$$T = T_2 + T_1 + T_0$$

$$T_2 = \frac{1}{2} \sum_{ij=1}^n m_{ij} \dot{q}_i \dot{q}_j$$

$$m_{ij} = \sum_{p=1}^n m_p \frac{\partial \vec{r}_p}{\partial q_i} \cdot \frac{\partial \vec{r}_p}{\partial q_j}$$

(m_{ij} is a function of $(q_1, \dots, q_n)$
because the derivatives $\frac{\partial \vec{r}_p}{\partial q_k}$
depend on $(q_1, \dots, q_n)$)

$$T_1 = \sum_{j=1}^n \pi_j \dot{q}_j$$

$$\pi_j = \sum_{p=1}^n m_p \frac{\partial \vec{r}_p}{\partial q_j} \cdot \frac{\partial \vec{r}_p}{\partial t}$$

$$T_0 = \sum_{p=1}^N \frac{m_p}{2} \left(\frac{\partial \vec{r}_p}{\partial t} \right)^2$$

Assuming $\frac{\partial \vec{r}_p}{\partial t} = \vec{0}$, and expanding for $q_j = q_{0j} + \eta_j$ with η_j small, gives

$$T = T_2 = \frac{1}{2} \sum_{ij=1}^n \left(m_{ij} \Big|_{\vec{q}=\vec{q}_0} + \sum_k \frac{\partial m_{ij}}{\partial q_k} \Big|_{\vec{q}=\vec{q}_0} \eta_k + \dots \right) (q_{0i} + \eta_i)' (q_{0j} + \eta_j)'$$

But $(q_{0i} + \eta_i)' = \dot{\eta}_i$ AND $\eta_k \dot{\eta}_i \dot{\eta}_j \propto \eta^3$ IS OF HIGHER ORDER IN SMALL DEVIATIONS

THEREFORE,
UP TO QUADRATIC
ORDER IN $\eta, \dot{\eta}$:

$$T = \frac{1}{2} \sum_{ij=1}^n T_{ij} \dot{\eta}_i \dot{\eta}_j \quad \text{with} \quad T_{ij} \equiv m_{ij} \Big|_{\vec{q}=\vec{q}_0}$$

$$\text{Det} [V_{ij} - \omega_k^2 T_{ij}] = 0 \quad k=1, \dots, n \quad n \text{ \# of degrees of freedom}$$

$$[V - \omega_k^2 \Pi] \vec{a}_k = \vec{0} \quad k=1, \dots, n$$

General solution is linear combination:

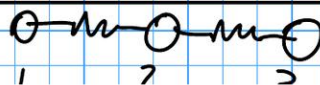
$$y_i = \sum_k C_k a_{ik} \cos(\omega_k t + \delta_k) \quad \vec{y} = \sum_k C_k \vec{a}_k \cos(\omega_k t + \delta_k)$$

Define $(A)_{ij} \equiv a_{ij}$ $A \equiv (\vec{a}_1 \vec{a}_2 \dots \vec{a}_n)$ Property: $1 = A^T A \Rightarrow A^{-1} = A^T \Pi$

$$A^T V A = \begin{pmatrix} \omega_1^2 & 0 & \dots & 0 \\ 0 & \ddots & & \\ 0 & & \omega_n^2 & \\ 0 & & & \omega_n^2 \end{pmatrix}$$

Matrix of eigenvector / normal modes

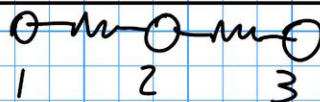
Normal modes example:



$$\mathbf{A}^\dagger \mathbf{V} \mathbf{A} = \begin{pmatrix} \omega_1^2 & 0 & 0 \\ 0 & \omega_2^2 & 0 \\ 0 & 0 & \omega_3^2 \end{pmatrix}$$

Matrix of eigenvector / normal modes

Normal modes example:



$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\ominus \rightarrow \ominus \leftarrow \ominus$$

ω_1

$$\vec{a}_2 = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

$$\leftarrow \ominus \rightarrow \leftarrow \ominus$$

ω_2

$$\vec{a}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\ominus \rightarrow \ominus \rightarrow \ominus$$

$\omega_3 = 0$

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3)$$

$$\vec{a}_j = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \omega_1 \quad \vec{0} \rightarrow \vec{0} \rightarrow \vec{0} \rightarrow \omega_3 = 0$$

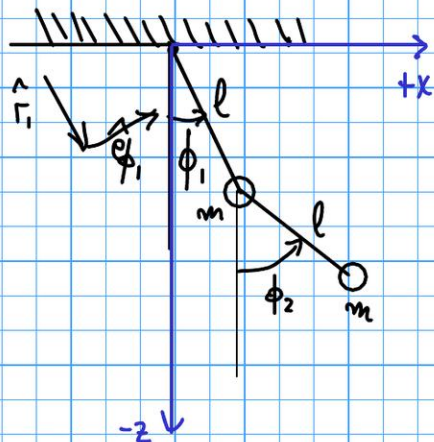
$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3)$$

Normal coordinates, ξ : $\vec{\eta} = A \vec{\xi} \quad \eta_i = \sum_k \xi_k a_{ik} \quad \xi_k = C_k \cos(\omega_k t + \delta_k) \quad \vec{\xi} = A^{-1} \vec{\eta} = A^{-1} \Pi \vec{\eta}$

$$L = \frac{1}{2} \sum_{ij} (\dot{\eta}_i T_{ij} \eta_j - \eta_i V_{ij} \eta_j) = \frac{1}{2} \left[\left(\frac{d\vec{\eta}}{dt} \right)^T T \vec{\eta} - \vec{\eta}^T V \vec{\eta} \right] = \frac{1}{2} \left[\dot{\vec{\xi}}^T A^T T A \vec{\xi} - \vec{\xi}^T A^T V A \vec{\xi} \right]$$

$$L = \frac{1}{2} \left[\left(\frac{d\vec{\xi}}{dt} \right)^T \vec{\xi} - \vec{\xi}^T \begin{pmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{pmatrix} \vec{\xi} \right] \Rightarrow \boxed{L = \frac{1}{2} \sum_{k=1}^n (\dot{\xi}_k^2 - \omega_k^2 \xi_k^2)}$$

Example: Oscillations of a double pendulum



Transformation equations

$$\vec{r}_1 = l (\sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}) = l \hat{r}_1$$

$$\vec{r}_2 = l (\sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}) + l (\sin \phi_2 \hat{x} - \cos \phi_2 \hat{z}) = l (\hat{r}_1 + \hat{r}_2)$$

$$\hat{r}_1 = \sin \phi_1 \hat{x} - \cos \phi_1 \hat{z} \quad \hat{r}_2 = \sin \phi_2 \hat{x} - \cos \phi_2 \hat{z}$$

Velocities

$$\dot{\hat{r}}_1 = \dot{\phi}_1 \frac{\partial \hat{r}_1}{\partial \phi_1}$$

$$\frac{\partial \hat{r}_1}{\partial \phi_1} = \hat{e}_{\phi_1} = \cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}$$

$$\dot{\hat{r}}_2 = \dot{\phi}_2 \frac{\partial \hat{r}_2}{\partial \phi_2}$$

$$\frac{\partial \hat{r}_2}{\partial \phi_2} = \hat{e}_{\phi_2} = \cos \phi_2 \hat{x} + \sin \phi_2 \hat{z}$$

$$\dot{\vec{r}}_1 = l \dot{\hat{r}}_1 = l \dot{\phi}_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{z})$$

$$\dot{\vec{r}}_2 = l (\dot{\hat{r}}_1 + \dot{\hat{r}}_2) = l \dot{\phi}_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}) + l \dot{\phi}_2 (\cos \phi_2 \hat{x} + \sin \phi_2 \hat{z})$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{z}^2) + m l^2 \dot{\phi}_1^2 + m l^2 \dot{\phi}_2^2 + 2 m l \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_1 - \phi_2)$$

$$z = z_1 + z_2 = l \phi_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{y}) + l \phi_2 (\cos \phi_2 \hat{x} + \sin \phi_2 \hat{y})$$

$$T = \frac{m}{2} \left[\left(\dot{\vec{r}}_1 \right)^2 + \left(\dot{\vec{r}}_2 \right)^2 \right] = \frac{m}{2} \left[l^2 \dot{\phi}_1^2 + l^2 (\dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \hat{e}_{\phi_1} \cdot \hat{e}_{\phi_2}) \right] \quad \hat{e}_{\phi_1} \cdot \hat{e}_{\phi_2} = \cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2 = \cos(\phi_2 - \phi_1)$$

$$T = \frac{m l^2}{2} [2 \dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1)] \quad V = m g z_1 + m g z_2 = -m g l (2 \cos \phi_1 + \cos \phi_2)$$

$$L = T - V = \frac{m l^2}{2} [2 \dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1)] + m g l (2 \cos \phi_1 + \cos \phi_2)$$

We know $\phi_1 = \phi_2 = 0$ is stable equilibrium position, but let's check

$$\frac{\partial V}{\partial \phi_1} = m g l \sin \phi_1 = 0 \text{ at } \phi_1 = 0 \quad \frac{\partial V}{\partial \phi_2} = m g l \sin \phi_2 = 0 \text{ at } \phi_2 = 0$$

And V is absolute minimum at $\phi_1 = \phi_2 = 0 \Rightarrow$ Equil is STABLE $\phi_i^0 = 0$

We thus expand T and V to quadratic order in $\eta_i = \phi_i - \phi_i^0 = \phi_i$

$$\frac{\partial V}{\partial \phi_1} = mgl \sin \phi_1 = 0 \text{ at } \phi_1 = 0 \quad \checkmark \quad \frac{\partial V}{\partial \phi_2} = mgl \sin \phi_2 = 0 \text{ at } \phi_2 = 0 \quad \checkmark$$

And V is absolute minimum at $\phi_1 = \phi_2 = 0 \Rightarrow$ Equil is STABLE

We thus expand T and V to quadratic order in $\eta_i = \phi_i - \phi_i^0 \stackrel{\phi_i^0=0}{=} \phi_i$

$$\dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1) = \dot{\eta}_1 \dot{\eta}_2 \cos(\eta_2 - \eta_1) \underset{\text{small } \eta_i}{\approx} \dot{\eta}_1 \dot{\eta}_2 \left[1 - \frac{(\eta_2 - \eta_1)^2}{2} + \dots \right] \approx \dot{\eta}_1 \dot{\eta}_2 + \mathcal{O}(\dot{\eta}^2 \eta^2)$$

$$\cos \phi_i = \cos \eta_i = 1 - \frac{\eta_i^2}{2} + \frac{\eta_i^4}{24} - \dots \approx 1 - \frac{\eta_i^2}{2}$$

Thus, to quadratic order in η and $\dot{\eta}$ we get

|| CHECK

$$T = \frac{1}{2} (\dot{\eta}_1 \dots \dot{\eta}_n) \mathbb{T} \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_n \end{pmatrix}$$

$$\cos \phi_i = \cos \eta_i = 1 - \frac{\eta_i^2}{2} + \cancel{\frac{\eta_i^4}{24}} - \dots \approx 1 - \frac{\eta_i^2}{2}$$

Thus, to quadratic order in η and $\dot{\eta}$ we get

$$V = -mgl \left[3 - \frac{2\eta_1^2}{2} - \frac{\eta_2^2}{2} \right] \Rightarrow V = mgl \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

$$T = \frac{ml^2}{2} [2\dot{\eta}_1^2 + \dot{\eta}_2^2 + 2\dot{\eta}_1\dot{\eta}_2] \Rightarrow T = ml^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

CHECK

$$(\eta_1 \ \eta_2) \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = (\eta_1 \ \eta_2) \begin{pmatrix} 2\eta_1 \\ \eta_2 \end{pmatrix} = 2\eta_1^2 + \eta_2^2 \quad \checkmark$$

$$(\dot{\eta}_1 \ \dot{\eta}_2) \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \end{pmatrix} = (\dot{\eta}_1 \ \dot{\eta}_2) \begin{pmatrix} 2\dot{\eta}_1 + \dot{\eta}_2 \\ \dot{\eta}_1 + \dot{\eta}_2 \end{pmatrix} = 2\dot{\eta}_1^2 + 2\dot{\eta}_1\dot{\eta}_2 + \dot{\eta}_2^2 \quad \checkmark$$

$$T = \frac{1}{2} (\dot{\eta}_1 \dots \dot{\eta}_n) \Pi \begin{pmatrix} \eta_1 \\ \vdots \\ \eta_n \end{pmatrix}$$

To simplify the algebra, we write $mg\ell = ml^2\left(\frac{g}{\ell}\right)$ $\omega_0^2 \equiv g/\ell$

Eigenvalue equation: $0 = \text{Det}[V - \omega^2 \Pi] = \text{Det}\left(ml^2 \begin{bmatrix} 2g/\ell - 2\omega^2 & -\omega^2 \\ -\omega^2 & g/\ell - \omega^2 \end{bmatrix}\right)$

$$\Leftrightarrow 0 = (2\omega_0^2 - 2\omega^2)(\omega_0^2 - \omega^2) - \omega^4 = 2\omega^4 - 4\omega^2\omega_0^2 + 2\omega_0^4 - \omega^4 = \omega^4 - 4\omega_0^2\omega^2 + 2\omega_0^4$$

$$\Leftrightarrow 0 = (\omega^2 - 2\omega_0^2)^2 - \underbrace{(4\omega_0^4 - 2\omega_0^4)}_{2\omega_0^4} \Leftrightarrow \omega^2 = 2\omega_0^2 \pm \sqrt{2}\omega_0^2 = (2 \pm \sqrt{2})\omega_0^2$$

COMPLETE SQUARES $\nearrow$

$$\omega = \begin{cases} \omega_4 = \sqrt{2+\sqrt{2}}\omega_0 \\ \omega = \sqrt{2-\sqrt{2}}\omega_0 \end{cases}$$

Eigenfrequencies

Eigenvector:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\omega_0^2 - 2\omega^2 & -\omega^2 \\ -\omega^2 & \omega_0^2 - \omega^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \Leftrightarrow \begin{cases} 0 = 2(\omega_0^2 - \omega^2)a_1 - \omega^2 a_2 \Leftrightarrow a_2 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)a_1 \\ 0 = -\omega^2 a_1 + (\omega_0^2 - \omega^2)a_2 \Leftrightarrow a_1 = \left(\frac{\omega_0^2}{\omega^2} - 1\right)a_2 \end{cases} \Leftrightarrow \begin{cases} \frac{a_2}{a_1} = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \\ \frac{a_2}{a_1} = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \end{cases}$$

SIDE NOTE: The 2 equations are equivalent $\Leftrightarrow 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \Leftrightarrow 1 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)\left(\frac{\omega_0^2}{\omega^2} - 1\right)$

$$\Leftrightarrow \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \Leftrightarrow 0 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) - \omega^4 \Leftrightarrow 0 = \text{Det}(V - \omega^2 \Pi) \quad \checkmark$$

are equivalent $\lambda^2 = (\omega^2) \quad \left(\frac{\omega_0^2}{\omega^2} - 1\right) = \frac{1}{\omega^2} (\omega^2 - 1)(\omega^2)$

$$\Leftrightarrow \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \Leftrightarrow 0 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) - \omega^4 \Leftrightarrow 0 = \text{Det}(V - \omega^2 \Pi) \quad \checkmark$$

so we can take either equation and write it with normalization to

$$\Leftrightarrow \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \Leftrightarrow 0 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) - \omega^4 \Leftrightarrow 0 = \text{Det}(V - \omega^2 \Pi) \quad \checkmark$$

so we can take either equation and use it, together with normalization, to get $\vec{a}$:

Propose $\vec{a} = N \begin{pmatrix} 1 \\ 2(\frac{\omega_0^2}{\omega^2} - 1) \end{pmatrix}$ N : normalization constant $x = \frac{\omega_0^2}{\omega^2}$

$$1 = [\vec{a}]^T \Pi [\vec{a}] = N^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} m \ell^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2(x-1) \end{pmatrix} = N^2 m \ell^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} \begin{pmatrix} 2x \\ 2x-1 \end{pmatrix}$$

$$1 = N^2 m \ell^2 [2x + 2(x-1)(2x-1)] = 2 N^2 m \ell^2 [2x^2 - 3x + x + 1] = 2 N^2 m \ell^2 [2x^2 - 2x + 1]$$

$$1 = 2 N^2 m \ell^2 [2(x-1)^2 + 4x - 2 + 1] = 2 N^2 m \ell^2 \left[\underbrace{2(x-1)^2 - 1}_{0} + 2x \right] = 4 N^2 m \ell^2 \frac{\omega_0^2}{\omega^2} \Rightarrow \boxed{N_{\pm} = \frac{1}{\sqrt{4m}} \frac{\omega_{\pm}}{\ell \omega_0} = \left(\frac{2 \pm \sqrt{2}}{4m \ell^2} \right)^{1/2}}$$

$$(x-1)^2 = x^2 - 2x + 1 \quad x^2 = (x-1)^2 + 2x - 1$$

$$\omega^4 [2(x-1)^2 - 1] = [2(x\omega^2 - \omega_0^2)^2 - \omega^4] = [2(\omega_0^2 - \omega^2)^2 - \omega^4] = 0$$

$$2x^2 - 2x + 1 = 2[(x-1)^2 + 2x - 1] - 2x + 1 = [2(x-1)^2 - 1] + 2x$$

$$\frac{\omega_0^2}{\omega_{\pm}^2} = \frac{1}{2 \pm \sqrt{2}} = \frac{2 \mp \sqrt{2}}{4 - 2} = 1 \mp \frac{1}{\sqrt{2}} \quad 2 \left[\frac{\omega_0^2}{\omega_{\pm}^2} - 1 \right] = \mp \sqrt{2}$$

$$\boxed{\vec{a}_{\pm} = \left(\frac{2 \pm \sqrt{2}}{4m \ell^2} \right)^{1/2} \begin{pmatrix} 1 \\ \mp \sqrt{2} \end{pmatrix}} \quad \leftrightarrow \quad \boxed{\omega_{\pm} = (2 \pm \sqrt{2})^{1/2} \frac{q}{\ell}}$$

Modal $\left[\begin{array}{c|c} A & \Rightarrow \Rightarrow \dots \Rightarrow \sqrt{3}/2 \quad b \end{array} \right] \quad a = (2 + \sqrt{2})^{1/2} = -c/\sqrt{2} \quad c = -\sqrt{2}a = -\sqrt{2}(2 + \sqrt{2})^{1/2}$

$$\omega_{\pm}^2 = \frac{2 \pm \sqrt{2}}{4 - 2} = 1 \mp \frac{1}{\sqrt{2}}$$

$$\sqrt{\omega_{\pm}^2 - 1} = \pm \frac{1}{\sqrt{2}}$$

$$\omega_{\pm}^2 = (4m\ell^2) \left(\mp \frac{1}{\sqrt{2}} \right)$$

$$\omega_{\pm}^2 = 1 \mp \frac{1}{\sqrt{2}}$$

Modal
matrix:

$$A = \begin{pmatrix} \vec{a}_+ & \vec{a}_- \end{pmatrix} = (4m\ell^2)^{\frac{1}{2}} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

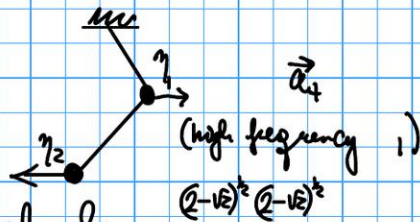
$$a = (2 + \sqrt{2})^{\frac{1}{2}} = -c/\sqrt{2}$$

$$c = -\sqrt{2} a = -\sqrt{2} (2 + \sqrt{2})^{\frac{1}{2}}$$

$$b = (2 - \sqrt{2})^{\frac{1}{2}} = d/\sqrt{2}$$

$$d = +\sqrt{2} b = +\sqrt{2} (2 - \sqrt{2})^{\frac{1}{2}}$$

Graphical representation:

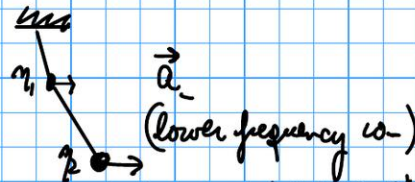


Normal coordinates:

$$\vec{\eta} = A^T \Pi \vec{\eta} = A^T m l^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$\vec{\eta} = (4m l^2)^{1/2} \begin{pmatrix} a & c \\ b & d \end{pmatrix} m l^2 \begin{pmatrix} 2\eta_1 + \eta_2 \\ \eta_1 + \eta_2 \end{pmatrix} = \left(\frac{m l^2}{4} \right)^{1/2} \begin{pmatrix} (2a+c)\eta_1 + (a+c)\eta_2 \\ (2b+d)\eta_1 + (b+d)\eta_2 \end{pmatrix}$$

$$\eta_+(t) = m^{1/2} l \left(\frac{16-1}{2\sqrt{2}} \right)^{1/2} \left[\sqrt{2}\eta(t) - \eta_2(t) \right] \quad \text{But we} \quad \eta_+(t) = F_+ \cos(\omega_+ t + \delta_+)$$



$$\begin{aligned} 2a+c &= (2-\sqrt{2})a = \sqrt{2}(\sqrt{2}-1)^2(\sqrt{2}-1)^2(2+\sqrt{2})^{1/2} = \sqrt{2} \frac{1}{4}(\sqrt{2}-1)^2 \\ a+c &= (1-\sqrt{2})a = -(\sqrt{2}-1)^2(\sqrt{2}-1)^2(2+\sqrt{2})^{1/2} = -\frac{1}{4}(\sqrt{2}-1)^2 \\ 2b+d &= (2+\sqrt{2})b = \sqrt{2}(\sqrt{2}+1)^2(\sqrt{2}+1)^2(2-\sqrt{2})^{1/2} = \sqrt{2} \frac{1}{4}(\sqrt{2}+1)^2 \\ b+d &= (1+\sqrt{2})b = (\sqrt{2}+1)^2(\sqrt{2}+1)^2(2-\sqrt{2})^{1/2} = \frac{1}{4}(\sqrt{2}+1)^2 \end{aligned}$$

$\left(\eta_2 = \sqrt{2} |\eta_1| > |\eta_1| \right)$
IN BOTH MODES

$$\begin{aligned} \left(\frac{m l^2}{4} \right)^{1/2} &= m^{1/2} \frac{l}{2} \\ \frac{1}{2} \frac{1}{4} (\sqrt{2}+1)^2 &= \left(\frac{16-1}{2\sqrt{2}} \right)^{1/2} \end{aligned}$$

$$\eta = \frac{1}{\sqrt{2}} \begin{pmatrix} b & d \end{pmatrix} \begin{pmatrix} \eta_1 + \eta_2 \\ \eta_1 - \eta_2 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} (2b+d)\eta_1 + (b+d)\eta_2 \\ (2b-d)\eta_1 + (b-d)\eta_2 \end{pmatrix}$$

$$\begin{aligned} \varphi_+(t) &= m \frac{k}{2} l \left(\frac{b-d}{2\sqrt{2}} \right)^2 \left[\sqrt{2} \eta_1(t) - \eta_2(t) \right] \\ \varphi_-(t) &= m \frac{k}{2} l \left(\frac{b+d}{2\sqrt{2}} \right)^2 \left[\sqrt{2} \eta_1(t) + \eta_2(t) \right] \end{aligned}$$

But we
know
also that:

$$\begin{aligned} \varphi_+(t) &= F_+ \cos(\omega_+ t + \delta_+) \\ \varphi_-(t) &= F_- \cos(\omega_- t + \delta_-) \end{aligned}$$

$$\begin{aligned} \dot{\varphi}_+(t) &= m \frac{k}{2} l \left(\frac{b-d}{2\sqrt{2}} \right)^2 \left[\sqrt{2} \dot{\eta}_1(t) - \dot{\eta}_2(t) \right] \\ \dot{\varphi}_-(t) &= m \frac{k}{2} l \left(\frac{b+d}{2\sqrt{2}} \right)^2 \left[\sqrt{2} \dot{\eta}_1(t) + \dot{\eta}_2(t) \right] \end{aligned}$$

$$\begin{aligned} \dot{\varphi}_+(t) &= -f_+ \omega_+ \sin(\omega_+ t + \delta_+) \\ \dot{\varphi}_-(t) &= -f_- \omega_- \sin(\omega_- t + \delta_-) \end{aligned}$$

Lagrangian in terms of the normal coordinates:

$$L = \frac{1}{2} \sum_{\alpha} (\dot{\varphi}_{\alpha}^2 - \omega_{\alpha}^2 \varphi_{\alpha}^2) = \frac{1}{2} \left[(\dot{\varphi}_+^2 - \omega_+^2 \varphi_+^2) + (\dot{\varphi}_-^2 - \omega_-^2 \varphi_-^2) \right]$$

$$L = \frac{1}{2} \left[\dot{\varphi}_+^2 - \left[(2+b) \frac{g}{2} \right] \varphi_+^2 \right] + \frac{1}{2} \left[\dot{\varphi}_-^2 - \left[(2-b) \frac{g}{2} \right] \varphi_-^2 \right]$$

$$L = \frac{1}{2} \sum_{\alpha} (\dot{\psi}_{\alpha} - \omega_{\alpha} \psi_{\alpha}) = \frac{1}{2} [(\dot{\psi}_{+} - \omega_{+} \psi_{+}) + (\dot{\psi}_{-} - \omega_{-} \psi_{-})]$$

$$L = \frac{1}{2} \left[\left(\dot{\psi}_{+}^2 - \left[(Q + \frac{g}{2}) \right] \psi_{+}^2 \right) + \left(\dot{\psi}_{-}^2 - \left[(Q - \frac{g}{2}) \right] \psi_{-}^2 \right) \right]$$

Assume initial condition is that at $t=0$: $\phi_1 = \phi_2 = 0$ $\dot{\phi}_1 = -\dot{\phi}_2 = \nu > 0$

$$L = \frac{1}{2} [(y_+ - L \cos \phi_+)^2 + (y_- - L \cos \phi_-)^2]$$

Assume initial condition is that at $t=0$: $\phi_1 = \phi_2 = 0$ $\dot{\phi}_1 = -\dot{\phi}_2 = v > 0$

Get $\phi_1(t), \phi_2(t)$

Then

$$f_+ \cos(\delta_+) = y_+(0) = 0 \Rightarrow \delta_+ = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots \quad m^{\frac{1}{2}} l \left(\frac{g \pm 1}{2l} \right)^{\frac{1}{2}} (g \pm 1) v = \left(\frac{g \pm 1}{2l} \right)^{\frac{1}{2}} m^{\frac{1}{2}} l v = \sqrt{2 \pm \frac{1}{2} g} \frac{m^{\frac{1}{2}} l v}{2}$$

$$f_- \cos(\delta_-) = y_-(0) = 0 \Rightarrow \delta_- = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$$

$$\begin{aligned} -f_+ \omega_+ \sin(\delta_+) &= \dot{y}_+(0) = m^{\frac{1}{2}} l \left(\frac{g-1}{2l} \right)^{\frac{1}{2}} [\sqrt{2} v - (-v)] = \sqrt{2 + \frac{1}{2} g} \frac{m^{\frac{1}{2}} l v}{2} \\ -f_- \omega_- \sin(\delta_-) &= \dot{y}_-(0) = m^{\frac{1}{2}} l \left(\frac{g+1}{2l} \right)^{\frac{1}{2}} [\sqrt{2} v + (-v)] = \sqrt{2 - \frac{1}{2} g} \frac{m^{\frac{1}{2}} l v}{2} \end{aligned}$$

Choose $\delta_+ = \delta_- = -\frac{\pi}{2}$

$$\Rightarrow f_{\pm} \sqrt{2 \pm \frac{1}{2} g} \sqrt{\frac{g}{2}} = \sqrt{2 \pm \frac{1}{2} g} \frac{m^{\frac{1}{2}} l v}{2} \Rightarrow f_{\pm} = \frac{v}{2} \left[\frac{m l^2}{g} \right]^{\frac{1}{2}}$$

$$\cos\left(\omega t - \frac{\pi}{2}\right) = \cos \omega t \cancel{\cos \frac{\pi}{2}} + \sin \omega t \left(\sin \frac{\pi}{2} \right) = \sin \omega t$$

$$\varphi_{\pm}(t) = f_{\pm} \cos(\omega_{\pm} t - \frac{\pi}{2}) = f_{\pm} \sin(\omega_{\pm} t)$$

$$\varphi_{\pm}(t) = \frac{1}{2} \left[\frac{m l^2}{g} \right]^{\frac{1}{2}} \sin \left(\left[(2 \pm \sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right)$$

$$a \equiv (2+\sqrt{2})^{\frac{1}{2}} = \frac{1}{\sqrt{2}} (\sqrt{2}+1)^{\frac{1}{2}} \quad c = -\sqrt{2} a$$

$$b \equiv (2-\sqrt{2})^{\frac{1}{2}} = \frac{1}{\sqrt{2}} (\sqrt{2}-1)^{\frac{1}{2}} \quad d = +\sqrt{2} b$$

$$(4 m l^2)^{-\frac{1}{2}} \frac{1}{2} \left[\frac{m l^2}{g} \right]^{\frac{1}{2}} = \frac{1}{4 \sqrt{2} g}$$

$$f_+ a = \frac{1}{2} \left[\frac{m l^2}{g} \right]^{\frac{1}{2}} (2+\sqrt{2})^{\frac{1}{2}}$$

$$f_- b = \frac{1}{2} \left[\frac{m l^2}{g} \right]^{\frac{1}{2}} (2-\sqrt{2})^{\frac{1}{2}}$$

In terms of the original coordinates

$$\vec{\eta}(t) = A \vec{\varphi}(t) = (4 m l^2)^{\frac{1}{2}} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \varphi_+(t) \\ \varphi_-(t) \end{pmatrix} = (4 m l^2)^{\frac{1}{2}} \begin{pmatrix} a \varphi_+(t) + b \varphi_-(t) \\ c \varphi_+(t) + d \varphi_-(t) \end{pmatrix}$$

$$\phi_1(t) = \eta_1(t) = \frac{1}{4 \sqrt{2} g} \left\{ (2+\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) + (2-\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\}$$

$$\phi_2(t) = \eta_2(t) = \frac{\sqrt{2}}{4 \sqrt{2} g} \left\{ -(2+\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) + (2-\sqrt{2})^{\frac{1}{2}} \sin \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\}$$

Check initial condition:

At $t=0$, all $\sin(\dots)$ terms are zero, so we have:

$$\phi_1 = \phi_2 = 0 \quad \checkmark$$

Check initial condition: At $t=0$, all $\cos(\dots)$ terms are 1, so we have:

$$\dot{\eta}_1(0) = \dot{\eta}_2(0) = \frac{1}{4 \sqrt{2} g} \left\{ (2+\sqrt{2})^{\frac{1}{2}} \cos \left(\left[(2+\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) - (2-\sqrt{2})^{\frac{1}{2}} \cos \left(\left[(2-\sqrt{2}) \frac{g}{l} \right]^{\frac{1}{2}} t \right) \right\}$$

$$\phi_2(t) = \psi_2(t) - \frac{1}{4\sqrt{2}} \left\{ (2+\sqrt{2}) \cos\left(\left[(2+\sqrt{2})\frac{g}{\hbar}\right]^{\frac{1}{2}} t\right) + (2-\sqrt{2}) \cos\left(\left[(2-\sqrt{2})\frac{g}{\hbar}\right]^{\frac{1}{2}} t\right) \right\}$$

$$\begin{aligned} \dot{\phi}_1(t) &= \dot{\eta}_1(t) = \frac{v}{4} \left\{ (2+\sqrt{2}) \cos\left(\left[(2+\sqrt{2})\frac{g}{\hbar}\right]^{\frac{1}{2}} t\right) + (2-\sqrt{2}) \cos\left(\left[(2-\sqrt{2})\frac{g}{\hbar}\right]^{\frac{1}{2}} t\right) \right\} \\ \dot{\phi}_2(t) &= \dot{\eta}_2(t) = \frac{\sqrt{2}v}{4} \left\{ -(2+\sqrt{2}) \cos\left(\left[(2+\sqrt{2})\frac{g}{\hbar}\right]^{\frac{1}{2}} t\right) + (2-\sqrt{2}) \cos\left(\left[(2-\sqrt{2})\frac{g}{\hbar}\right]^{\frac{1}{2}} t\right) \right\} \end{aligned}$$

Check initial condition: At $t=0$, all $\cos(\dots)$ terms are 1, so we have:

$$\dot{\phi}_1 = \frac{v}{4} \{ (2+\sqrt{2}) + (2-\sqrt{2}) \} = v \quad \checkmark$$

$$\dot{\phi}_2 = \frac{\sqrt{2}v}{4} \{ -(2+\sqrt{2}) + (2-\sqrt{2}) \} = -v \quad \checkmark$$

COMPARISON WITH QUANTUM MECHANICS:

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Q M

$$|\psi(0)\rangle \rightarrow \{c_k\}$$

$$|\psi(t)\rangle \leftarrow c_k e^{-iE_k t/\hbar}$$

$$|\psi(0)\rangle = \sum_k c_k |\varphi_k\rangle \quad c_k = \langle \varphi_k | \psi(0) \rangle$$

Small osc

$$\vec{q}(0), \vec{p}(0) \longrightarrow \vec{q}(0), \dot{\vec{q}}(0)$$

$$\vec{q}(t), \dot{\vec{q}}(t) \longleftarrow \vec{q}(t), \dot{\vec{q}}(t)$$

$$\vec{q}(t) = A \vec{q}(t) \quad \dot{\vec{q}}(t) = A^T \dot{\vec{q}}(t)$$

$$|\psi(0)\rangle = \sum_{\alpha} c_{\alpha} |\varphi_{\alpha}\rangle \quad c_{\alpha} = \langle \varphi_{\alpha} | \psi(0) \rangle$$

$$|\psi(t)\rangle = \sum_{\alpha} c_{\alpha} e^{-iE_{\alpha}t/\hbar} |\varphi_{\alpha}\rangle$$

$$\vec{\eta}(t), \vec{\eta}(t) \leftarrow \vec{\xi}(t), \vec{\xi}(t)$$

$$\vec{\eta}(t) = A \vec{\xi}(t) \quad \vec{\xi}(t) = A^{-1} \vec{\eta}(t)$$